Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 1 · Write polynomials in standard form

1. **Simplify.** $(3c^3 - c + 11) - (c^2 + 2c + 8)$

A) $3c^3 - c^2 - 3c + 3$

B) $3c^3 + c^2 + c + 19$

C) $3c^3 - c^2 - c + 3$

D) $2c^3 - 3c + 3$

2. **Simplify.** $(2x - 2y + 1) - (3y + 4x)$

A) $6x + y + 1$

B) $-2x - 5y + 1$

C) $-2x - y + 1$

D) $-2x - 5y - 1$

Section 2 · Multiply binomials by using the FOIL method

3. **Find the product.** $(2y - 11)(y^2 - 3y + 2)$

A) $2y^3 - 17y^2 + 37y - 22$

B) $2y^3 - 6y^2 + 4y - 22$

C) $2y^3 + 17y^2 + 37y - 22$

D) $2y^3 - 17y^2 - 37y + 22$

4. **Find the product.** $(4a + 7)(9a^2 + 2a - 7)$

A) $36a^3 + 71a^2 - 14a - 49$

B) $36a^3 + 8a^2 - 28a - 49$

C) $36a^3 - 55a^2 - 14a - 49$

D) $13a^3 + 71a^2 - 14a - 49$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 3 · Factor trinomials of the form x^2+bx+c

5. **Factor the polynomial, if possible.** $5x^2 + 34x + 24$

A) $(5x + 4)(x + 6)$ **B)** $(5x + 6)(x + 4)$

C) $(5x + 12)(x + 2)$ **D)** prime

6. **Factor the polynomial, if possible.** $2x^2 + 3x + 6$

A) $(2x + 3)(x + 2)$ **B)** $(2x + 2)(x + 3)$

C) $(2x + 6)(x + 1)$ **D)** prime

7. **Factor the polynomial, if possible.** $5x^2 + 3x + 4$

A) $(5x + 4)(x - 1)$ **B)** $(5x + 1)(x + 4)$

C) $(5x + 2)(x + 2)$ **D)** prime

Section 4 · Factor perfect square trinomials

8. **Find a value for** m **that makes** $4x^4 - 44x^2 + m$ **a perfect square trinomial, then factor it.**

A) $m = 121;\ (2x^2 - 11)^2$

B) $m = 11;\ (2x^2 - 11)^2$

C) $m = 121;\ (2x^2 + 11)^2$

D) $m = 22;\ (2x^2 - 11)(2x^2 + 11)$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 5 · Find and use the sum of the measures of the interior angles of a polygon

9. **In quadrilateral QRST the interior angles are $Q = (2x + 5)^\circ$, $R = x^\circ$, $S = (2x + 7)^\circ$, $T = x^\circ$. Find the measure of each angle.**

A) $121^\circ, 58^\circ, 123^\circ, 58^\circ$

B) $120^\circ, 60^\circ, 124^\circ, 56^\circ$

C) $58^\circ, 121^\circ, 58^\circ, 123^\circ$

D) $121^\circ, 58^\circ, 121^\circ, 58^\circ$

Section 6 · Recognize and apply the properties of the sides and angles of parallelograms

10. **Quadrilateral ABCD is a rectangle. If $m\angle ADB = (4x + 8)^\circ$ and $m\angle DBA = (6x + 12)^\circ$, find x.**

A) $x = 5$

B) $x = 7$

C) $x = 8$

D) $x = 9$

Section 7 · Recognize the conditions that ensure a quadrilateral is a parallelogram

11. **In a quadrilateral, one pair of opposite sides is both parallel and congruent. Which statement is correct?**

A) Yes; it is a parallelogram (a pair of opposite sides is parallel and congruent)

B) No; both pairs of opposite sides must be parallel

C) Only if the diagonals are congruent

D) Only if all four sides are congruent

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 7 · Recognize the conditions that ensure a quadrilateral is a parallelogram

12. **Both pairs of opposite angles of a quadrilateral are congruent. Is it a parallelogram?**

A) No

B) Yes; both pairs of opposite angles are congruent

C) Only if it is a rectangle

D) Only if the diagonals bisect each other

Section 8 · Recognize and apply the properties of kites

13. **In trapezoid ABCD, $\overline{AD} \parallel \overline{BC}$ with $A(-3, 2),\ B(-3, -3),\ C(3, 0),\ D(1, 4)$. Find the endpoints of the midsegment.**

A) $(-3, -0.5)$ and $(2, 2)$

B) $(-3, 2)$ and $(3, 0)$

C) $(0, 2)$ and $(2, 2)$

D) $(-3, -0.5)$ and $(2, -2)$

Section 9 · Find the scale factor of a dilation

14. **Triangle J'K'P' is the image of JKP. $J(2, 4),\ K(4, 4)$ and $J'(4, 8),\ K'(8, 8)$. Find the scale factor.**

A) $\frac{1}{2}$

B) 2

C) 4

D) $\frac{1}{4}$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 10 · Solve problems using the properties of similar polygons

15. Two polygons are similar. Corresponding sides give $\frac{x+1}{8} = \frac{3x+1}{20}$. Find x.

A) $x = 2$

B) $x = 3$

C) $x = 4$

D) $x = \frac{12}{5}$

Section 11 · Find the scale factor of a dilation

16. A'B'C'D' is the image of ABCD under a dilation. $C(6, -6)$, $D(-6, -6)$ and $C'(4, -4)$, $D'(-4, -4)$. Find the scale factor.

A) $\frac{3}{2}$

B) $\frac{2}{3}$

C) 2

D) $\frac{1}{3}$

17. A figure and its dilated image are drawn on the coordinate plane. Two matching points: original $(-2, 0)$ and $(-2, 3)$; image $(-5, 0)$ and $(-5, 7.5)$. Find the scale factor.

A) 1.5

B) 2.5

C) 2

D) 0.4

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 11 · Find the scale factor of a dilation

18. **A dilation with center the origin and scale factor $k = 0.5$ is applied. Find the image of $K(-4, 4)$.**

A) $K'(-8, 8)$

B) $K'(-2, 2)$

C) $K'(-4, 2)$

D) $K'(2, -2)$

Section 12 · Identify similar triangles using the AA Similarity Postulate

19. **In △RST, ∠S = 42° and ∠T = 98°. In △UVW, ∠V = 98° and ∠U = 40°. Are the triangles similar?**

A) Yes; $\triangle RST \sim \triangle UWV$ by AA

B) No; the 42° angle has no match

C) Yes, by SSS

D) Not enough information

Section 13 · Use proportional parts within triangles

20. **In △ADC, point E lies on AD and point B lies on AC with $\overline{EB} \parallel \overline{DC}$. If AB = 6, BC = 4 and AE = 9, find ED.**

A) $ED = 3$

B) $ED = 6$

C) $ED = 13.5$

D) $ED = 2.7$

21. **In △ADC, point E lies on AD and point B lies on AC with $\overline{EB} \parallel \overline{DC}$. If AB = 12, AC = 16 and ED = 5, find AE.**

A) $AE = 10$

B) $AE = 15$

C) $AE = 20$

D) $AE = 3.75$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 14 · Recognize and use proportional relationships of corresponding angle bisectors, altitudes,…

22. **An angle bisector of a triangle divides the opposite side into 8 and 12. The sides next to those parts are m and 26. Find m to the nearest tenth.**

A) $m = 17.3$ **B)** $m = 39$

C) $m = 12$ **D)** $m = 21.7$

23. **An angle bisector divides a side into 7 and x. The sides next to those parts are 20 and 24 (7 goes with 20). Find x.**

A) $x = 8.4$ **B)** $x = 5.8$

C) $x = 11$ **D)** $x = 3.5$

Section 15 · Find the geometric mean between two numbers

24. **Find the geometric mean between 4 and 6.**

A) 5 **B)** $2\sqrt{6} \approx 4.9$

C) $\sqrt{10} \approx 3.2$ **D)** 24

25. **Find the geometric mean between $\frac{1}{2}$ and 2.**

A) 1 **B)** $\frac{5}{4}$

C) $\frac{1}{4}$ **D)** $\sqrt{2}$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 16 · Use the Pythagorean Theorem and its converse

26. **The legs of a right triangle are 3 and 3. Find the hypotenuse x.**

A) $x = 3\sqrt{2} \approx 4.2$

B) $x = 6$

C) $x = \sqrt{6} \approx 2.4$

D) $x = 18$

Section 17 · Use the properties of 45°-45°-90° triangles and use the properties of 30°-60°-90° triangles

27. **A square has diagonal 13. Find the side length x and the angle y that the diagonal makes with a side.**

A) $x = \frac{13\sqrt{2}}{2};\ y = 45^\circ$

B) $x = 13\sqrt{2};\ y = 45^\circ$

C) $x = \frac{13}{2};\ y = 30^\circ$

D) $x = \frac{13\sqrt{3}}{2};\ y = 60^\circ$

Section 18 · Use trigonometric ratios to find side lengths and angle measures of right triangles

28. **In a right triangle the hypotenuse is 40 and the side opposite angle y is 22. Find y to the nearest tenth.**

A) $y = 33.4^\circ$

B) $y = 56.6^\circ$

C) $y = 28.7^\circ$

D) $y = 61.3^\circ$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 19 · Use trigonometric ratios to find side lengths and angle measures of right triangles

29. **Carlos (6 ft tall) sights the top of a garage at a 42° angle of elevation. He steps back 20 ft and the angle is 10°. How tall is the garage to the nearest foot?**

A) 10 ft
B) 4 ft
C) 16 ft
D) 7 ft

Section 20 · Find the area of a triangle using two sides and an included angle

30. **A triangular piece of scenery has two sides of 8 ft and 10 ft with an included angle of 34°. Find its area to the nearest tenth.**

A) $22.4\ \text{ft}^2$
B) $44.8\ \text{ft}^2$
C) $33.2\ \text{ft}^2$
D) $40\ \text{ft}^2$

Section 21 · Factorize binomials that are the difference of squares and factor perfect square trinomials

31. **Factor the expression.** $9a^2 - 4b^2$

A) $(3a + 2b)(3a - 2b)$
B) $(3a - 2b)^2$
C) $(9a + 4b)(a - b)$
D) prime

32. **Factor the expression.** $16 - x^2$

A) $(4 - x)^2$
B) $(4 + x)(4 - x)$
C) $(x - 4)(x - 4)$
D) prime

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 22 · Find and use the sum of the measures of the exterior angles of a polygon

33. **The exterior angles of a hexagon measure $x°$, $2x°$, $(x + 10)°$, $(x + 18)°$, $3x°$, $(x - 1)°$. Find x.**

A) $x = 33$ **B)** $x = 37$

C) $x = 40$ **D)** $x = 60$

34. **The exterior angles of a quadrilateral measure $(x + 10)°$, $31°$, $(2x - 42)°$, $(x - 11)°$. Find x.**

A) $x = 93$ **B)** $x = 90$

C) $x = 79$ **D)** $x = 103$

Section 23 · Use similar triangles to solve problems

35. **ROOFING. In the roof outline $\triangle DEF \sim \triangle FBC$ by SAS with DF = 10 ft, FC = 16 ft and FB = 8 ft. Find DE = x.**

A) $x = 5$ **B)** $x = 6.4$

C) $x = 12.8$ **D)** $x = 4$

36. **RADIO. At the same time an 8-foot shadow is cast by a radio tower, a yardstick (36 in) casts a shadow 0.5 in long. How tall is the tower?**

A) 576 ft **B)** 16 ft

C) 144 ft **D)** 0.5 ft

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 24 · Find the distance between two points on the coordinate plane

37. Find the distance between $Y(-1, 0)$ and $Z(0, -1)$.

A) $\sqrt{2} \approx 1.41$

B) 2

C) $\sqrt{10} \approx 3.16$

D) 1

38. Find the distance between $X(-3, -2)$ and $Y(-1, 0)$.

A) $\sqrt{2} \approx 1.41$

B) $\sqrt{8} \approx 2.83$

C) $\sqrt{10} \approx 3.16$

D) 4

39. Find the distance between $X(-3, -2)$ and $Z(0, -1)$.

A) $\sqrt{8} \approx 2.83$

B) $\sqrt{10} \approx 3.16$

C) $\sqrt{2} \approx 1.41$

D) 4

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 25 · Determine whether three given measures of a triangle define 0, 1, or 2 triangles by using…

40. **In △ABC, ∠A = 15°, ∠C = 72° and AB = c = 375. Find BC = x to the nearest tenth.**

A) $x \approx 102.1$ **B)** $x \approx 95.0$

C) $x \approx 375$ **D)** $x \approx 1403.0$

41. **In a triangle, ∠A = 50°, ∠C = 68° and the side opposite C is 5. Find the side x opposite A to the nearest tenth.**

A) $x \approx 4.1$ **B)** $x \approx 6.7$

C) $x \approx 3.2$ **D)** $x \approx 5.0$